Enterpreneurial Mindset

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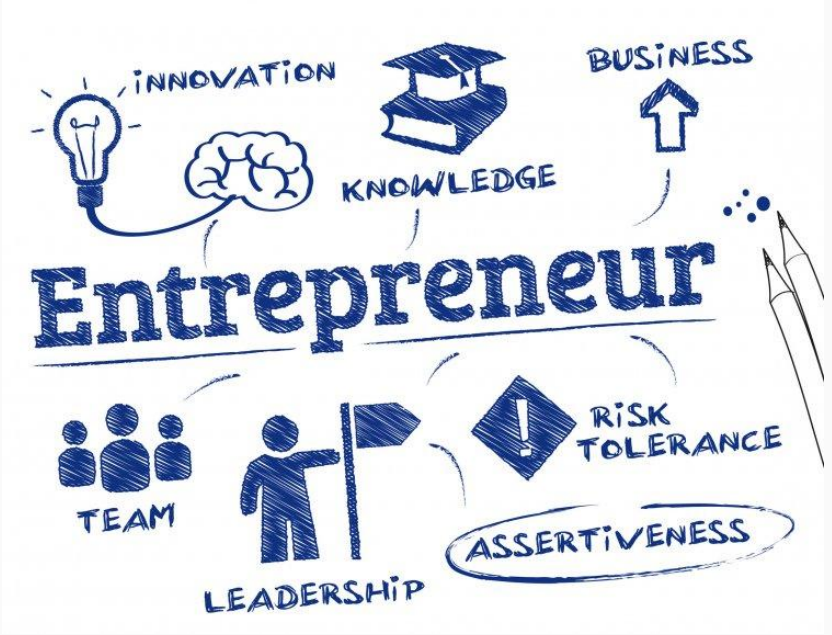
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Outline

- Entrepreneurial Mindset in Agile Projects (DSDM Context)
- Factors Affecting Software Development Costs
- Proven Techniques for Estimating Project Budgets
- Top-Down Estimation
- Bottom-Up Estimation

Entrepreneurial Mindset in Agile Projects (DSDM Context)

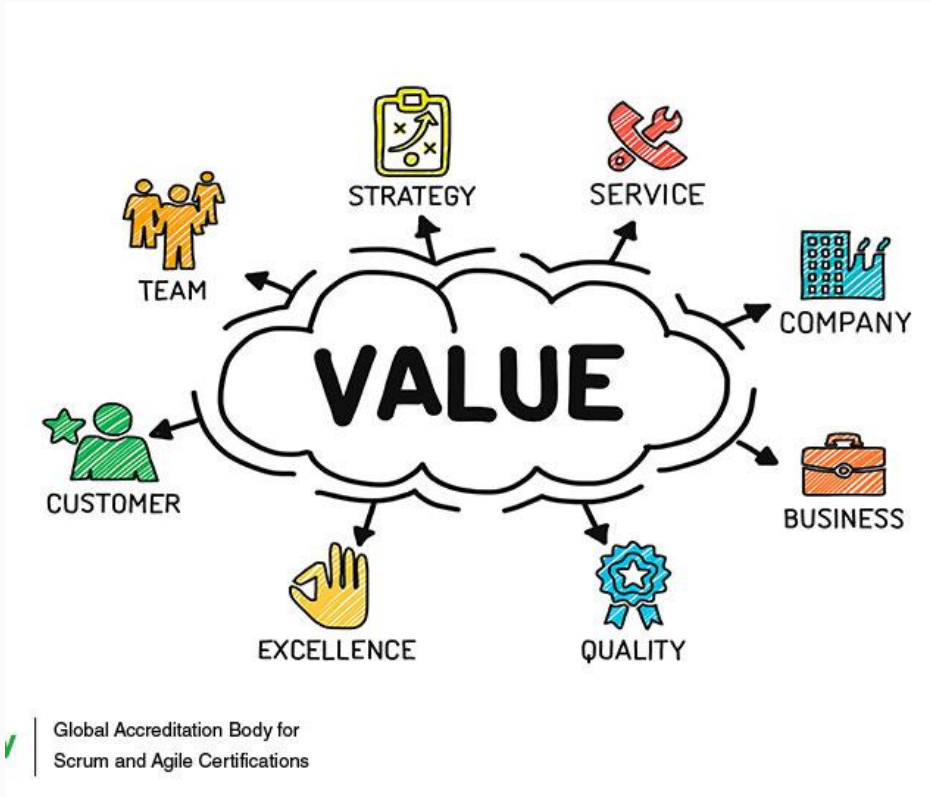


What is an Entrepreneurial Mindset?

An **entrepreneurial mindset** in Agile software projects focuses on creating real business value, understanding the market and users, managing risk proactively, and ensuring long-term viability of the product. Rather than measuring success only by technical completion, success is measured by whether the system delivers benefits that stakeholders are willing to support, fund, and use.

Key Characteristics:

- Value-driven decision making
- Customer and market awareness
- Risk-aware and opportunity-focused
- Balancing innovation with control



Entrepreneurial Mindset in Example of Software Project

Case: *University Smart Attendance & Analytics System*
A web-based system used by universities to manage attendance, generate analytics, and provide compliance reports for management.

1. Value-Driven Decision Making

Explanation:

Decisions are prioritised based on **business impact**, not just technical interest or feature quantity.

Case Example:

Instead of building advanced AI analytics first, the team prioritises:

- Automated attendance reporting for accreditation audits

This feature:

- Solves an urgent management problem
- Is classified as **Must Have** under MoSCoW
- Delivers immediate business benefit

➡️ *Entrepreneurial thinking ensures effort is spent on what stakeholders value most.*

Entrepreneurial Mindset in Example of Software Project

Case: *University Smart Attendance & Analytics System*
A web-based system used by universities to manage attendance, generate analytics, and provide compliance reports for management.

2. Strong Customer and Market Awareness

Explanation:

Teams understand **who the users are**, **what they need**, and **how the system fits into the market or organisation.**

Case Example:

Through stakeholder interviews, the team identifies:

- Lecturers want quick attendance entry
- Faculty management wants high-level compliance dashboards

As a result:

- Different user stories are created for lecturers and management
- The system is positioned as a **decision-support tool**, not just an attendance app

➡️ *The product is shaped by real user needs, increasing adoption and buy-in.*

Entrepreneurial Mindset in Example of Software Project

Case: *University Smart Attendance & Analytics System*
A web-based system used by universities to manage attendance, generate analytics, and provide compliance reports for management.

3. Risk-Aware and Opportunity-Focused Thinking

Explanation:

Entrepreneurial teams **anticipate risks early** while actively seeking opportunities to enhance value.

Case Example:

Identified risks:

- Data privacy concerns (student personal data)
- System downtime during peak academic periods

Actions taken:

- Early inclusion of authentication and access control
- Cloud hosting with uptime monitoring

At the same time, the team identifies an opportunity:

- Offering advanced analytics as a premium feature for university management

➡️ *Risks are controlled early while new opportunities are explored.*

Entrepreneurial Mindset in Example of Software Project

Case: *University Smart Attendance & Analytics System*
A web-based system used by universities to manage attendance, generate analytics, and provide compliance reports for management.

4. Balance Between Innovation, Cost, Security, and Quality

Explanation:

Entrepreneurial mindset avoids over-engineering by balancing **innovation with practicality.**

Case Example:

The team considers adding:

- Facial recognition for attendance

However, they decide:

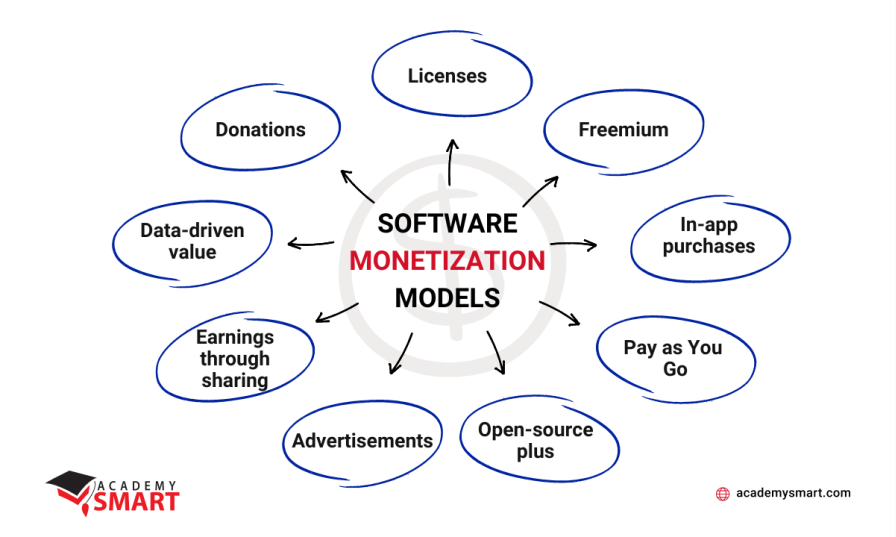
- High cost and privacy concerns outweigh benefits at this stage
- QR-based attendance with audit logs is sufficient and compliant

This decision:

- Reduces development cost
- Improves deployment speed
- Maintains acceptable security and quality levels

➡️ *Innovation is applied wisely, not blindly.*

Monetization Strategies



Why Monetization Matters?

Software projects must generate business value.

Common Monetization Strategies:
Subscription model – Monthly / yearly SaaS fees
Usage-based pricing – Per user, per transaction, or per API call
Licensing model – One-time or periodic licence fees
Freemium model – Free core features, paid premium services
Enterprise contracts – Customised solutions with SLA commitments

Agile / DSDM Link:
Influences MoSCoW prioritisation and cost–benefit decisions.

Monetization Strategies

1. Subscription Model (Monthly / Yearly SaaS Fees)

Explanation:

Users pay a recurring fee to access the system and its features.

Case Application:

Universities subscribe annually based on:

- Number of active students
- Number of faculties using the system

Example:

- Basic plan: Attendance recording + standard reports
- Premium plan: Advanced analytics + compliance dashboards

Agile / DSDM Impact:

- Subscription-critical features are prioritised as **Must Have**
- Enables predictable revenue for continuous improvement

2. Usage-Based Pricing (Per User / Transaction / API Call)

Explanation:

Customers pay based on actual system usage, making pricing flexible and scalable.

Case Application:

The university pays based on:

- Number of active lecturers using the system per semester
- Number of analytics reports generated by management

Example:

- RM1 per active lecturer per month
- RM5 per advanced analytics report

Agile / DSDM Impact:

- Encourages lean feature development
- Cost–benefit decisions guide which usage metrics are monetised

Monetization Strategies

3. Licensing Model (One-Time or Periodic Licence Fees)

Explanation:

Customers purchase a licence to deploy the system on their own infrastructure.

Case Application:

Large universities choose:

- On-premise deployment due to data governance policies

Example:

- One-time licence fee + annual maintenance fee
- Custom deployment for internal servers

Agile / DSDM Impact:

- Core system stability becomes **Must Have**
- Enhancements delivered incrementally through maintenance cycles

4. Freemium Model (Free Core, Paid Premium Features)

Explanation:

Basic functionality is free to attract users, while advanced features are paid.

Case Application:

- Small colleges use free attendance recording
- Paid upgrade unlocks analytics, export, and audit logs

Example:

- Free: Manual attendance entry
- Paid: Automated reports + accreditation-ready exports

Agile / DSDM Impact:

- Free features delivered early for adoption
- Premium features prioritised for revenue generation

Monetization Strategies

5. Enterprise Contracts (Customised Solutions with SLA)

Explanation:

Tailored solutions with contractual guarantees for large organisations.

Case Application:

Public universities sign enterprise contracts requiring:

- 99.9% uptime
- Dedicated support
- Data protection and compliance assurance

Example:

- Custom dashboards for Vice-Chancellor level
- SLA with response and resolution times

Agile / DSDM Impact:

- SLA-related features treated as **Must Have NFRs**
- Influences estimation, planning, and prioritisation

Market Research & Target Audience



Purpose of Market Research

To understand who the system is for, what problems it solves, and why users will adopt it.

Key Activities:

- Identify target users (students, lecturers, administrators, enterprises)
- Analyse user pain points and expectations
- Study competitors and alternative solutions
- Validate product–market fit early

Agile / DSDM Link:

Drives user stories and acceptance criteria
Supports iterative feedback and refinement
Reduces risk of delivering low-value features

Purpose of Market Research

1. Identify Target Users

Explanation:

Different stakeholders have different goals, workflows, and success criteria.

Case Application:

Market research identifies four main user groups:

- **Students** – Want quick, non-intrusive attendance marking
- **Lecturers** – Need fast class management and minimal admin work
- **Administrators** – Require accurate records for audits
- **University management** – Want dashboards for compliance and decision-making

Agile / DSDM Impact:

- Separate **user stories** are written for each role
- Features are prioritised based on stakeholder value (MoSCoW)

2. Analyse User Pain Points and Expectations

Explanation:

Understanding frustrations and expectations ensures the system solves real problems.

Case Application:

Findings from interviews and surveys:

- Lecturers find manual attendance time-consuming
- Administrators struggle to compile audit reports
- Management lacks real-time visibility of compliance status

Resulting features:

- One-click attendance recording
- Automated report generation
- Real-time compliance dashboards

Agile / DSDM Impact:

- Acceptance criteria reflect real operational needs
- High-impact features classified as **Must Have**

Purpose of Market Research

3. Study Competitors and Alternative Solutions

Explanation:

Analysing existing solutions helps teams identify gaps and opportunities.

Case Application:

The team reviews:

- Spreadsheet-based attendance systems
- Generic LMS attendance modules

Identified gaps:

- Limited analytics for management
- Poor audit and reporting capabilities

Differentiation strategy:

- Focus on analytics and compliance reporting
- Position system as a **management decision-support tool**

Agile / DSDM Impact:

- Competitive features prioritised early
- Avoids duplicating low-value features

4. Validate Product–Market Fit Early

Explanation:

Early validation confirms whether users and stakeholders find value in the system.

Case Application:

- A prototype dashboard is shown to faculty management
- Feedback highlights strong interest in compliance tracking

Actions taken:

- Dashboard features refined in early Timeboxes
- Less-used features deferred to later releases

Agile / DSDM Impact:

- Supports iterative feedback and refinement
- Reduces risk of investing effort in unwanted features

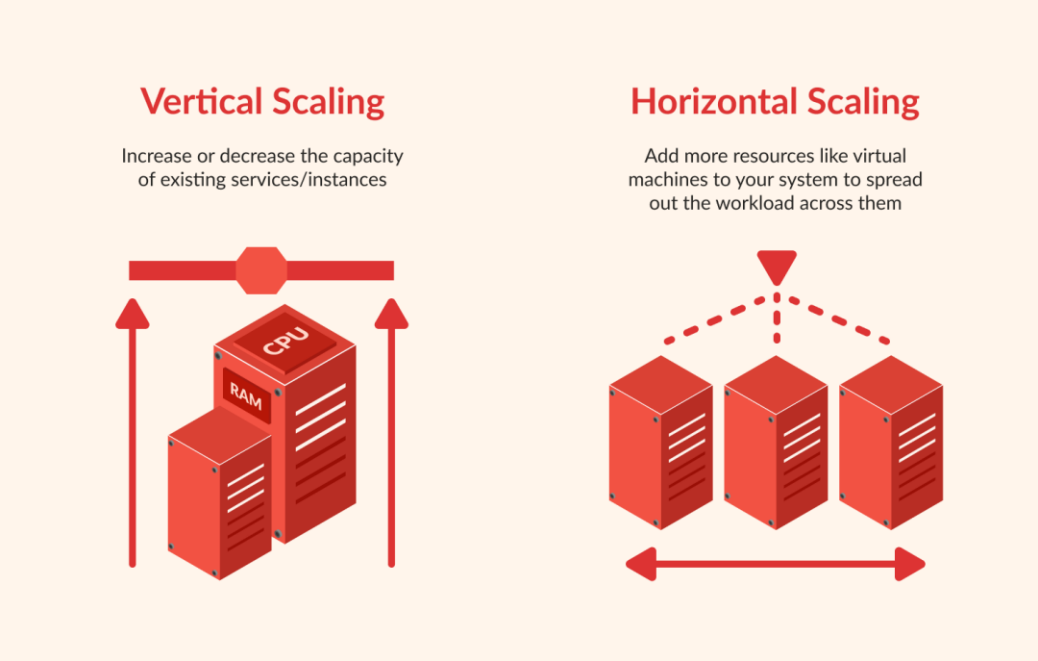
Scalability and Growth



Scalability and Growth

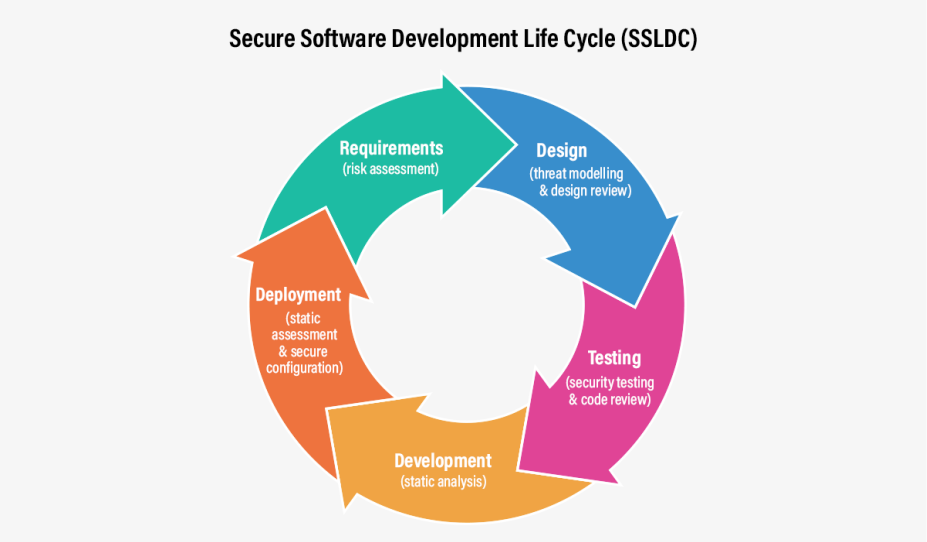
Why Scalability Matters:
Entrepreneurial systems are designed for future growth.

Scalability Considerations:
- Increasing number of users
- Growing data volume
- Performance under peak load
- Modular and extensible architecture



Agile / DSDM Link:
Captured through Non-Functional Requirements.

Security in Software Project Management



Security in Software Project Management

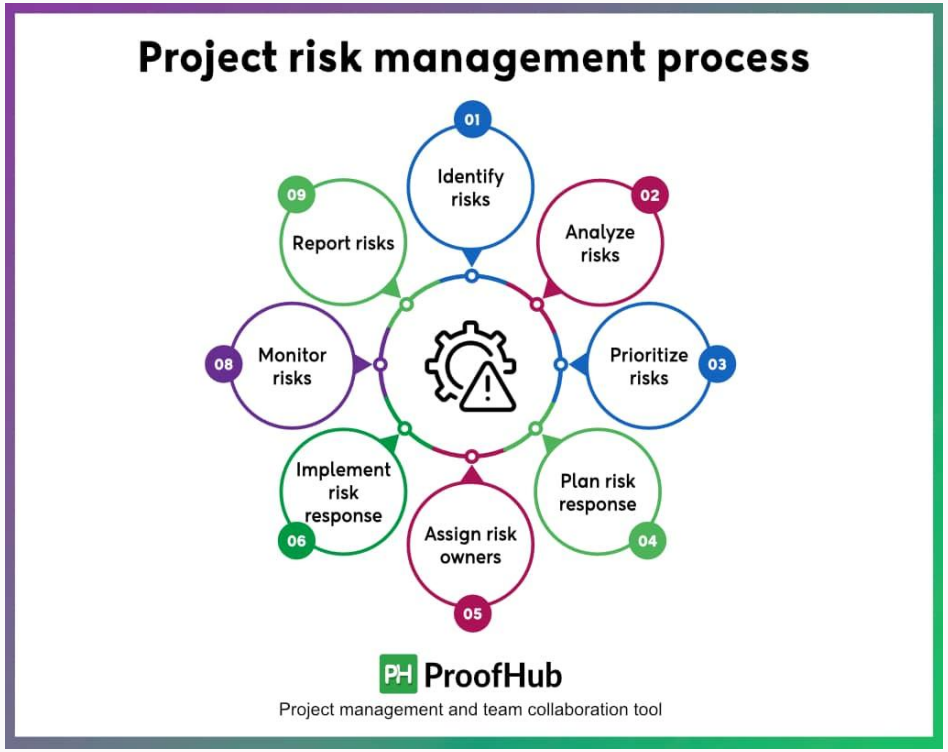
Security as a Business Enabler:
Security protects reputation, trust, and revenue.

Security Considerations:

- Data confidentiality and privacy
- Access control and authentication
- Regulatory compliance (e.g. GDPR, PDPA)
- Risk assessment and mitigation

Agile / DSDM Link:

- Security requirements treated as Must Have NFRs
- Managed continuously throughout the lifecycle



Security Considerations

1. Data Confidentiality and Privacy

Explanation:

Sensitive data must be protected from unauthorised access or exposure.

Case Application:

The system stores:

- Student personal information
- Attendance history
- Lecturer and staff identifiers

Security measures applied:

- Role-based data access
- Encryption of stored attendance records

Project Management Impact:

- Privacy requirements captured as **Must Have NFRs**
- Early risk identification prevents legal and reputational issues

2. Access Control and Authentication

Explanation:

Only authorised users should access specific system functions.

Case Application:

- Students can only view their own attendance
- Lecturers can manage attendance for their classes
- Administrators and management access aggregated reports

Security features:

- Login with institutional credentials
- Role-based access control (RBAC)

Project Management Impact:

- Access rules influence system architecture
- Included in early Timeboxes to reduce security risk

Security Considerations

3. Regulatory Compliance (e.g. GDPR, PDPA)

Explanation:

Compliance ensures the system adheres to data protection laws and institutional policies.

Case Application:

The system supports compliance by:

- Limiting data retention periods
- Providing audit logs for data access
- Allowing controlled data export for accreditation

Project Management Impact:

- Compliance requirements drive scope definition
- Treated as **non-negotiable Must Have requirements**

4. Risk Assessment and Mitigation

Explanation:

Security risks must be identified, analysed, and mitigated throughout the project lifecycle.

Case Application:

Identified risks:

- Data breaches during peak usage
- System downtime during exam periods

Mitigation actions:

- Secure hosting with uptime monitoring
- Regular backup and recovery planning

Project Management Impact:

- Risks reviewed at every Timebox review
- Prevents late-stage security rework

Service Level Agreement (SLA)



Service Level Agreement (SLA)

What is an SLA?
A formal agreement defining service expectations.

Typical SLA Elements:
- System availability
- Response and resolution times
- Performance benchmarks
- Security clauses
- Penalties for non-compliance

Agile / DSDM Link:
Influences estimation and prioritisation.



Example SLA (Case Study: University Smart Attendance & Analytics System)

SLA Section	SLA Clause (Excerpt)	Case Study Explanation (Why It Matters)
Service Parties	**Service Provider:** ABC Tech Solutions **Customer:** XYZ University	Defines contractual responsibility between the software vendor and university management
System Availability (Uptime)	System availability guaranteed at **99.5% per month**, excluding scheduled maintenance. Maintenance notified **48 hours in advance** and conducted outside peak hours.	Ensures attendance and reporting systems are available during teaching weeks, exams, and audits
Response & Resolution Times	**Critical:** Response ≤ 1 hr, Resolution ≤ 4 hrs **High:** Response ≤ 2 hrs, Resolution ≤ 8 hrs **Medium:** Response ≤ 8 hrs, Resolution ≤ 2 working days **Low:** Response ≤ 24 hrs	Guarantees fast support during critical academic periods and operational disruptions
Performance Benchmarks	Attendance submission response ≤ **3 seconds** (95% of requests). Analytics dashboard loads ≤ **5 seconds** under normal load.	Maintains smooth lecturer operations and timely management decision-making
Security & Data Protection	Data encrypted **in transit and at rest**. **Role-based access control** enforced. Compliance with **PDPA** and university data policies. Audit logs retained for **12 months**.	Protects sensitive student and staff data and supports compliance audits
Penalties for Non-Compliance	For uptime below SLA: **5% service credit** per 0.1% shortfall. Repeated breaches (3 months) allow contract review or termination.	Encourages service reliability and protects the university's operational and financial interests

Integration of Information Security

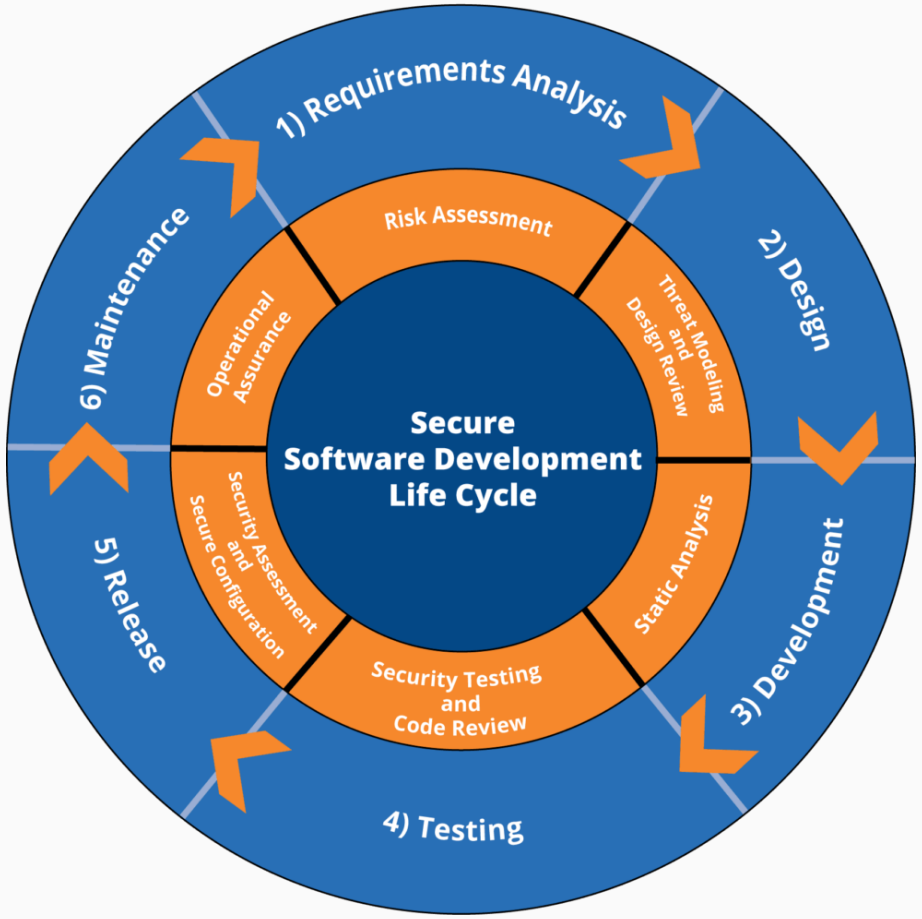


Integration of Information Security

Security by Design:
Security embedded throughout the lifecycle.

Key Practices:
- Secure architecture design
- Data encryption
- Secure APIs
- Regular security testing

Agile / DSDM Link:
Integrated into every Timebox.



Educating Team Members on Security



Educating Team Members on Security

Why Team Education Matters:
Human error is one of the largest security risks.

Education & Awareness Activities:
- Secure coding practices
- Awareness of common attacks
- Incident response procedures
- Compliance and ethics

Agile / DSDM Link:
Supports empowered, self-organising teams.



Use of Security Tools

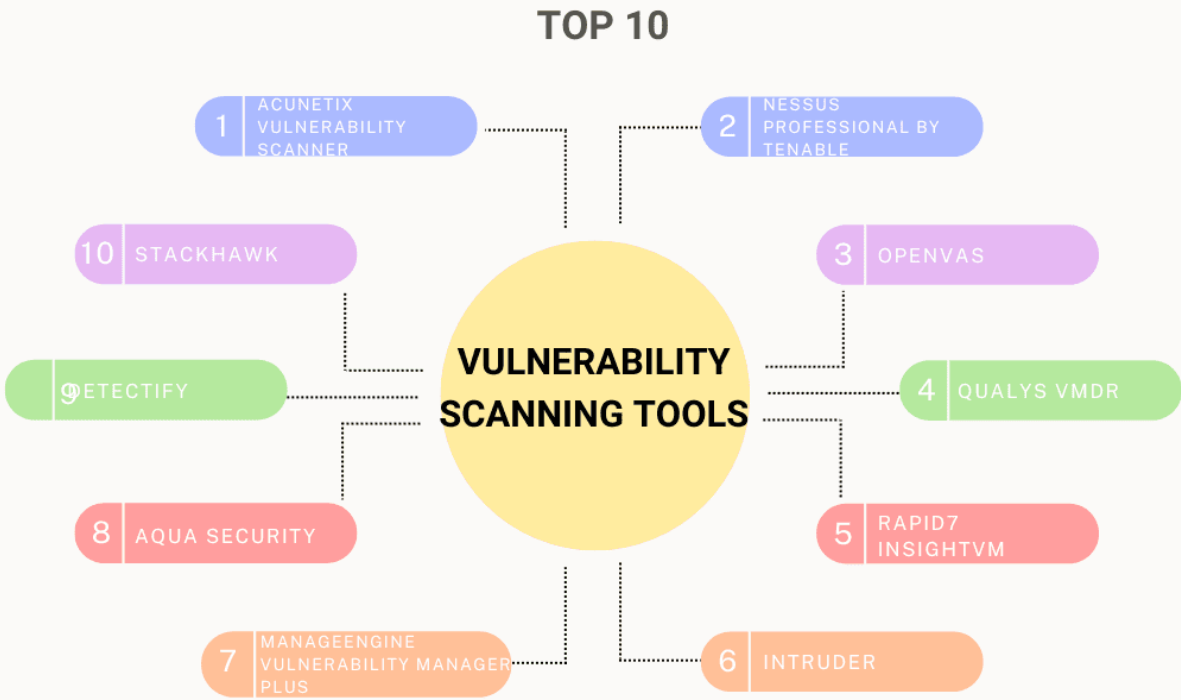


Use of Security Tools

Role of Security Tools:
Support automation and detection.

Examples of Security Tools:
- Static and dynamic code analysis
- Vulnerability scanners
- Identity and access management
- Logging and monitoring systems

Agile / DSDM Link:
Integrated into CI/CD pipelines.



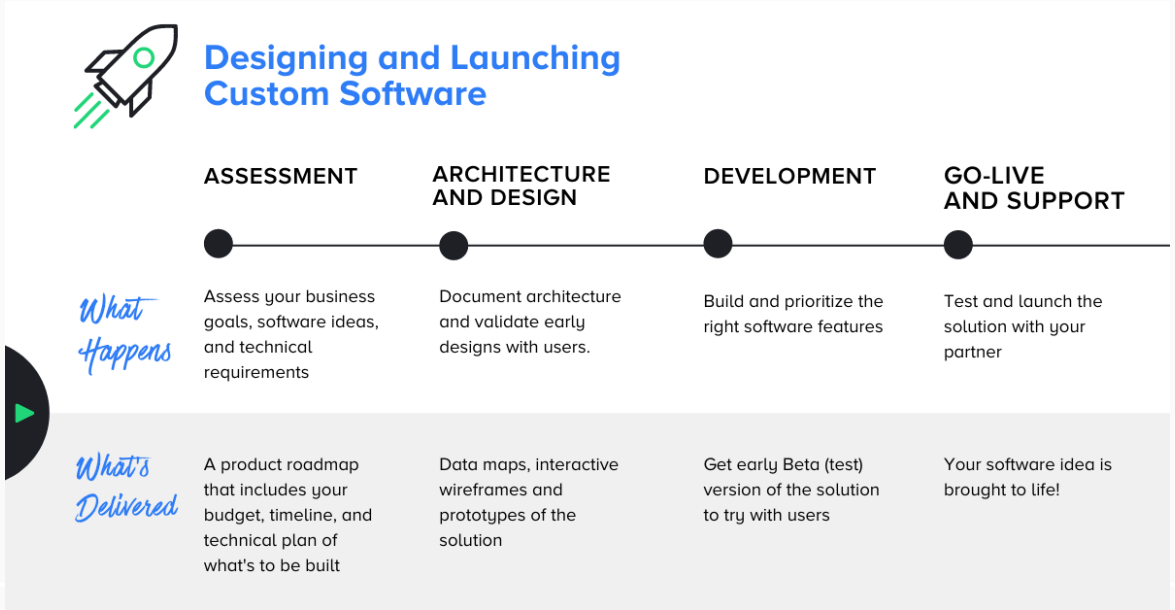
Summary: Entrepreneurial Mindset in Agile DSDM



Summary – Entrepreneurial Mindset in Agile DSDM

Key Takeaways:
• Entrepreneurial mindset aligns Agile delivery with business success
• Combines market awareness, revenue thinking, scalability, and security
• Supports DSDM principles of business focus, quality, and sustainability

DSDM Impact
Supports DSDM principles of:
- Business focus
- Built-in quality
- Continuous delivery
- Long-term sustainability





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